# 102: Linux

May, 2023

# Audience

- Those who have some basic knowledge of how an OS works

# Materials

- Filesystems overview
- Linux filesystem
- Bash command
- Shell script flow control
- Networking

US patent

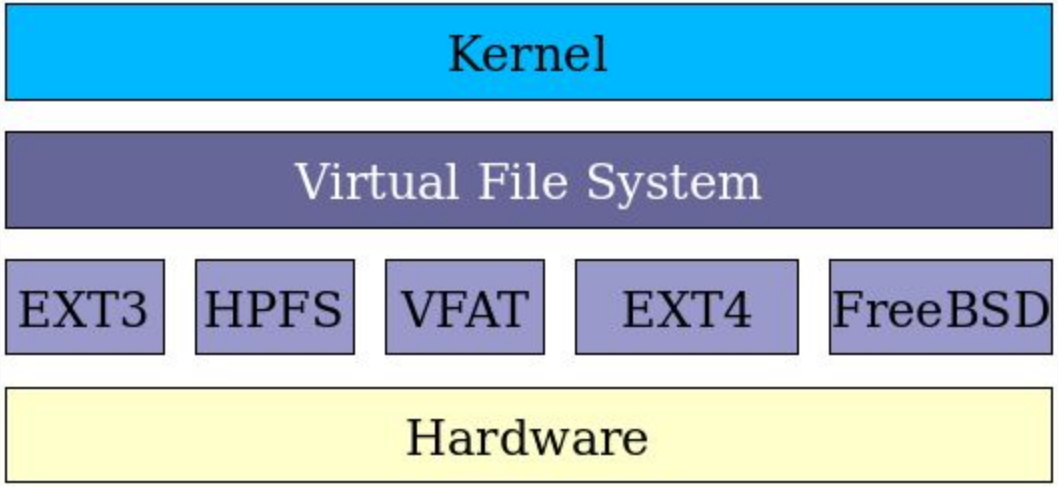
# File Allocation Table (FAT) (1977~)

- FAT12 (1982, MS-DOS 1.25)
  - Volume max 32MB
  - Max no. of files: 4068
- FAT16 (1988, MS-DOS 4.0)
  - Volume max 2GB*
  - Max no. of files: 65460
- FAT32 (1996, Windows 95 OSR2)
  - Max file size: 4GB
  - Volume max 16TB*
- exFAT (2006)
  - Max file size: 16EB*
  - Volume max 128PB
- 8.3 filename and LFN
  - FILENA~1.TXT
- NTFS (1993, Windows NT 3.1)
  - Volume max 256TB-8PB*
  - Max file size: 16EB
  - encryption, compression
  - journaling

# Linux and MacOS

- ext2/ext3/ext4
- btrfs
- XFS
- ZFS
- ...

- HFS/HFS+
- APFS

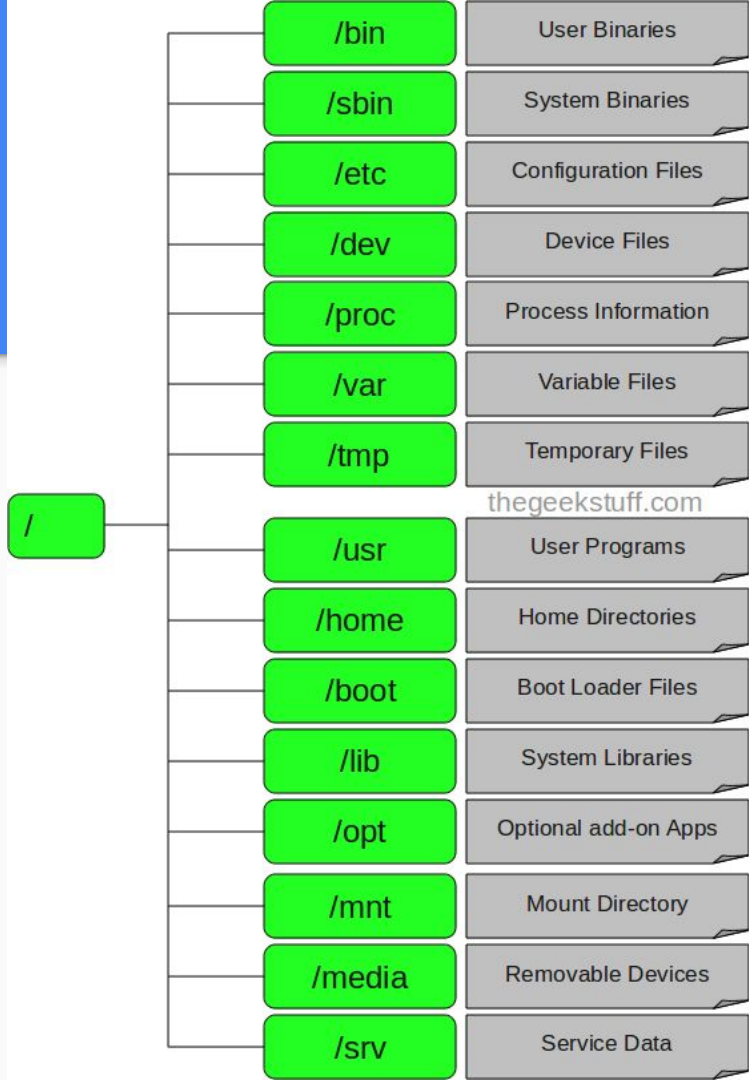


# Directory/folder

```
$ ls /
```

[Filesystem Hierarchy Standard](https://www.pathname.com/fhs/pub/fhs-2.3.pdf)

https://www.pathname.com/fhs/pub/fhs-2.3.pdf



# Partition vs volume

Drag to resize a volume.

Size of volume

Disk Utility

View | Volume | First Aid | Partition | Erase | Restore | Unmount | Info

**Partition device "APPLE SSD AP0512Q Media"?**

Click add (+) to increase the number of partitions on the device. Click on each partition to change the name and format. Drag the resize control or enter the size of each partition and click Apply.

Media
95 GB

Macintosh HD, Macint...edia, Special Projects
399 GB

**Device Information**

Device: APPLE SSD AP0512Q Media

Scheme: GUID Partition Map

Size: 500.28 GB

**Partition Information**

Name: Media

Format: APFS

Size: 95 GB

This partition will be created.

* Small partition shown larger than actual size.

? Revert | Cancel | Apply

Disk Management

File Action View Help

| Volume | Layout | Type | File System | Status | Capacity | Free Spa... | % Free |
|---|---|---|---|---|---|---|---|
| (H:) | Simple | Basic | NTFS | Healthy (P... | 192.08 GB | 180.56 GB | 94 % |
| (J:) | Simple | Basic | NTFS | Healthy (P... | 63.29 GB | 63.12 GB | 100 % |
| New Volume (F:) | Striped | Dynamic | NTFS | Healthy | 20.27 GB | 20.16 GB | 99 % |
| New Volume (G:) | Simple | Dynamic | NTFS | Healthy | 10.24 GB | 10.15 GB | 99 % |
| New Volume (I:) | Mirror | Dynamic | NTFS | Healthy | 1003.62 GB | 1002.25 ... | 100 % |
| System Reserved | Simple | Basic | NTFS | Healthy (S... | 100 MB | 64 MB | 64 % |
| System Reserved (... | Simple | Basic | NTFS | Healthy (A... | 628 MB | 376 MB | 60 % |

**Disk 0**
Basic
60.00 GB
Online

- System 100 MB Healthy
- (C:) 30.60 GB NTFS Healthy (Boot, Page
- (E:) 10.08 GB NTFS Healthy (Primary I
- 19.21 GB Unallocated

**Disk 1**
Dynamic
1024.00 GB
Online

- New Volume (G:) 10.24 GB NTFS Healthy
- New Volume (F:) 10.14 GB NTFS Healthy
- New Volume (I:) 1003.62 GB NTFS Healthy

**Disk 2**
Dynamic
2048.00 GB

- New Volume (F:) 10.14 GB NTFS
- New Volume (I:) 1003.62 GB NTFS
- 1034.24 GB

Unallocated | Primary partition | Simple volume | Striped volume | Mirrored volume

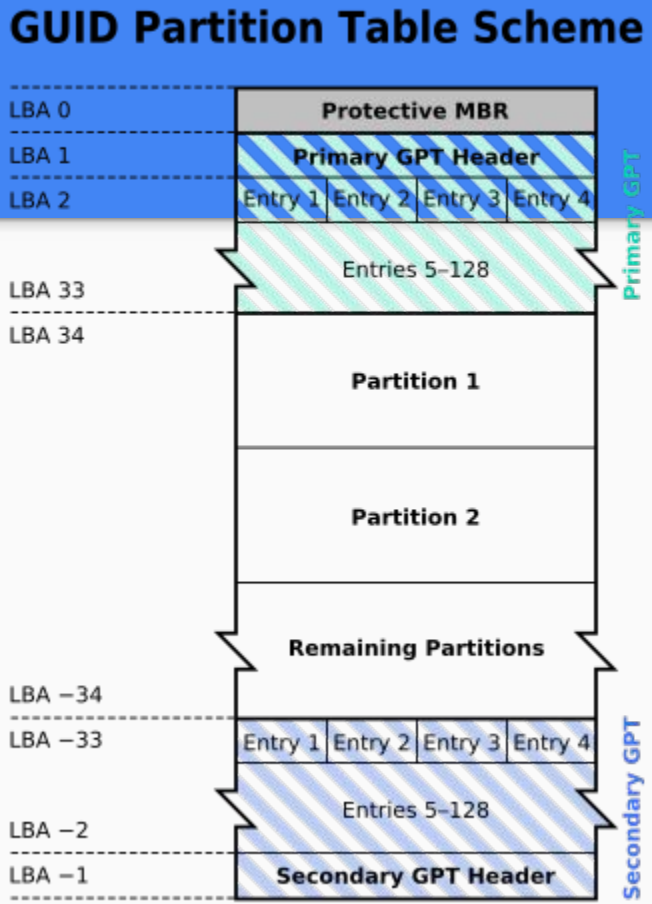
# Master Boot Record (MBR) (1983)

- Boot code
  - 16 bit
  - beginning of the disk
- Partition table
  - 4 primary partitions
  - or 3 primary partitions and 1 extended partition (many logical partitions inside)
  - max 2TB disk

1990s, Intel

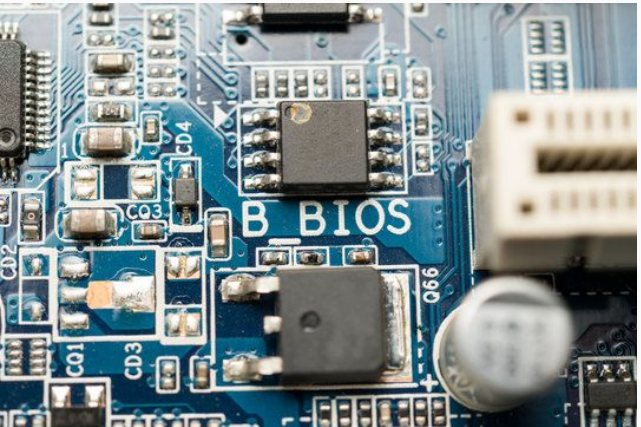
# GUID Partition Table (GPT)

- No boot code
- Stored at the beginning and the end of the disk
- Max 256 partitions
- Max 64ZB disk
- Windows Vista ~ requires UEFI



# BIOS vs UEFI

- BIOS (Basic Input/Output System) (1975)
  - initialize hardware, boot OS
  - 16 bit




- UEFI (Unified Extensible Firmware Interface) (2004)
  - initialize hardware, boot OS
  - 32/64 bit
  - support GPT
  - support mouse, network, FAT, (NTFS), secure boot*
  - boot from (ESP)/EFI/boot/bootx64.efi
  - support legacy boot using CSM

CMOS Setup Utility - Copyright (C) 1984-1999 Award Software

| | |
|---|---|
| ▶ Standard CMOS Features | ▶ Frequency/Voltage Control |
| ▶ Advanced BIOS Features | Load Fail-Safe Defaults |
| ▶ Advanced Chipset Features | Load Optimized Defaults |
| ▶ Integrated Peripherials | Set Supervisor Password |
| ▶ Power Management Setup | Set User Password |
| ▶ PnP/PCI Configurations | Save & Exit Setup |
| ▶ PC Health Status | Exit Without Saving |

Esc : Quit ↑ ↓ → ← : Select Item
F10 : Save & Exit Setup

Time, Date, Hard Disk Type...

Aptio Setup - AMI

Main | Advanced | AMD CBS | AMD PBS Option | Chipset | Server Mgmt | Security | Boot | Save & Exit

| | |
|---|---|
| BIOS Information | |
| Project Name | MZ72-HB0-00 |
| Project Version | M02 |
| Build Date and Time | 02/05/2021 18:22:06 |
| BMC Information | |
| BMC Firmware Version | 12.50.09 |
| Processor Information | |
| CPU 0 Brand String | AMD EPYC 7763 64-Core Processor |
| CPU 1 Brand String | AMD EPYC 7763 64-Core Processor |
| CPU Speed | 2450 MHz |
| Processor Core | 64 |
| Microcode Patch | A001114 |
| Total Memory | 524288 MB |
| Memory Speed | 3200 MT/s |
| VR Information | |
| Version | 8160 |
| AGESA PI Version | |
| PI Version | 1.0.0.0 |

→←: Select Screen
↑↓: Select Item
Enter: Select
+/-: Change Opt.
F1: General Help
F3: Previous Values
F9: Optimized Defaults
F10: Save & Exit
ESC: Exit

Version 2.21.1279 Copyright (C) 2021 AMI

msi CLICK BIOS 5

Advanced (F7) | F12 | English

16:31 Wed 4 Mar, 2020

GAME BOOST

HW

A-XMP

OFF

CPU Speed 3.90 GHz
DDR Speed 2666 MHz

CPU Temperature: 34°C
MotherBoard Temperature: 32°C

Boot Priority

MB: MEG X570 ACE (MS-7C35)
CPU: AMD Ryzen 7 3800X 8-Core Processor
Memory Size: 16384MB
VCore: 1.444V
DDR Voltage: 1.220V
BIOS Ver: E7C35AMS.180
BIOS Build Date: 01/16/2020

EZ Mode

- CPU
- Memory
- Storage
- Fan Info
- Help

Current DRAM Frequency:

2666 MHz

Current DRAM Size: 16384MB DRAM Voltage: 1.220V

| | Model | Size |
|---|---|---|
| DIMMA1: | unknown | N/A |
| DIMMA2: | A-DATA | 8192 MB |
| DIMMB1: | unknown | N/A |
| DIMMB2: | A-DATA | 8192 MB |

XMP Profile
Profile1: DDR4 3600MHz 17-18-18-38
Profile2: DDR4 2666MHz 16-16-16-39

The WindowsClub

- M-Flash
- Favorites
- Hardware Monitor

LAN Option ROM
ErP Ready
Indication LED Control ON

CSM (UEFI) ON
AHCI /RAID ON
RGB Light Control ON

# Install Linux: a demonstration